OFFICIAL ARCHITECTURE & OPERATIONAL MANUAL

Natural Milk Dairy

End-to-End Smart Dairy Subscription, Fleet Dispatch & Digital Credit Management Platform

APPLICATION VERSION **v2.4.0 (Production Ready)**	ARCHITECTURE PATTERN **Decoupled MERN + Realtime Bus**	TARGET DEPLOYMENT **Node.js / React / MongoDB**
PRIMARY PORTALS **Admin · Customer · Driver · Web**	LIVE STATE SYNC **BroadcastChannel & Polling Bus**	DOCUMENTATION DATE **October 2026**

Executive Summary

The **Natural Milk Dairy Platform** is a full-stack, enterprise-grade milk distribution and subscription system engineered to streamline farm-to-doorstep morning operations. It integrates 4 distinct user interfaces (Owner Console, Customer Self-Service Portal, Delivery Partner Driver App, and Public E-Commerce Web Experience) into a unified, high-availability architecture with automated delivery schedule generators, vacation pause & instant resume state machines, real-time driver cash reconciliation, and digital ledger accounting.

1. Technology Stack & System Architecture

The platform leverages a modern, robust, and highly scalable JavaScript/React ecosystem. Every component was engineered with performance, accessibility, zero third-party UI framework lock-in, and offline-tolerant mobile usability in mind.

Frontend Core & Client Layer

High-speed, component-driven client architecture optimized for desktop consoles and mobile web apps.

React 18.3 | Vite 5.4 | Lucide React 0.474 | Vanilla CSS Design System | BroadcastChannel API

Backend API & Micro-Services Layer

Modular Express server handling REST endpoints, ACID transaction sessions, and cron-like schedulers.

Node.js 24 Runtime | Express 4.21 | MongoDB & Mongoose 8.9 | JWT Auth (RFC 7519) | Bcryptjs Password Hashing

Cloud Media & Email Delivery

Automated CDN media management and transactional credential and invoice messaging.

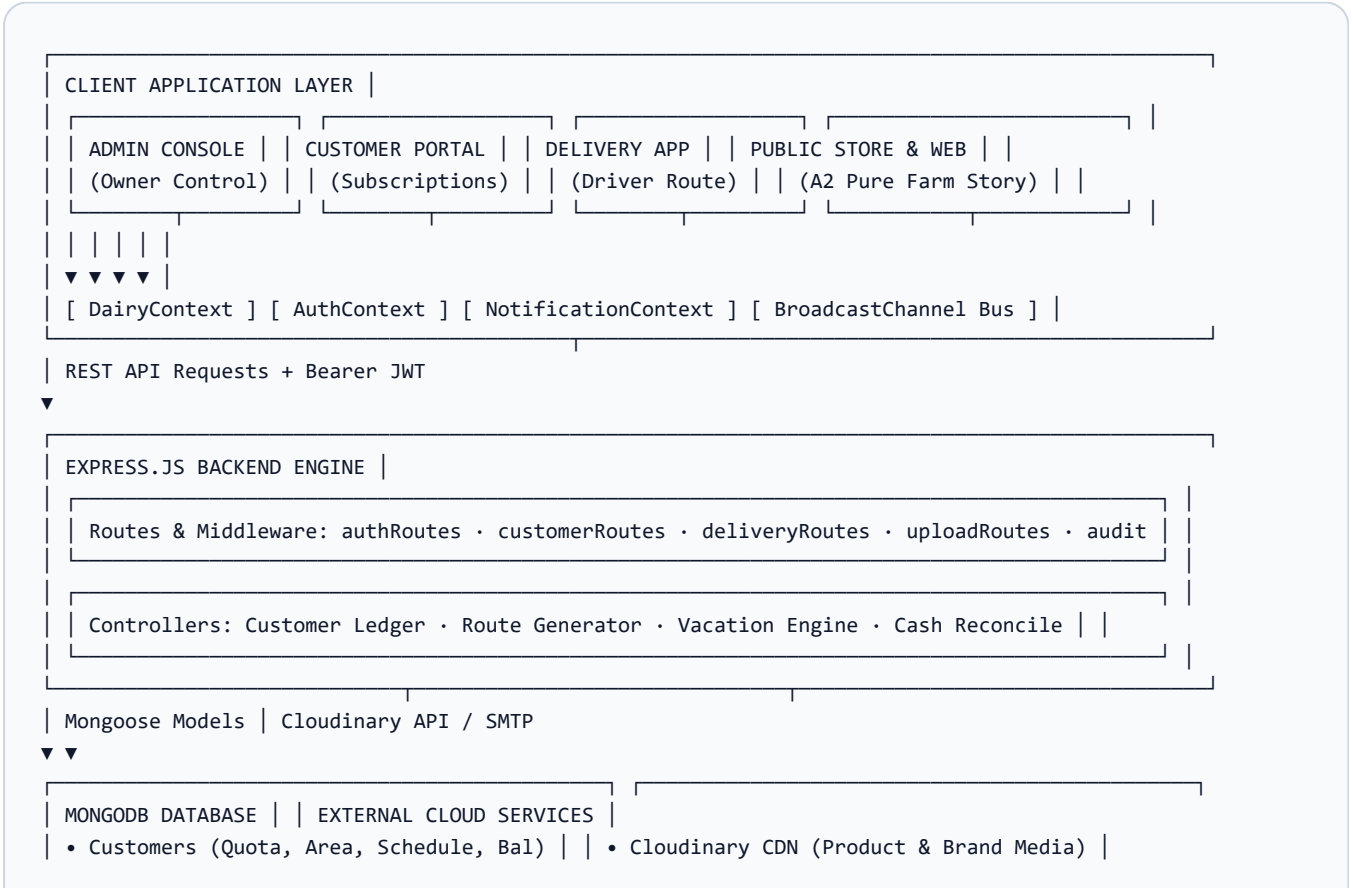
Cloudinary v2 SDK | Multer Multipart Uploads | Nodemailer 10.0 | Cloud WebP Delivery

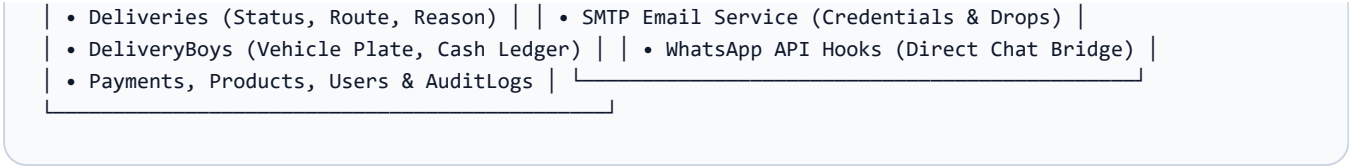
DevOps, Diagnostics & Monitoring

Zero-latency logging, structured error handlers, and audit trail record keeping.

Morgan HTTP Logger | Dotenv Config | AuditLog History Engine | Safe Session Rollbacks

System Architecture Diagram




• Deliveries (Status, Route, Reason)
• DeliveryBoys (Vehicle Plate, Cash Ledger)
• Payments, Products, Users & AuditLogs
• SMTP Email Service (Credentials & Drops)
• WhatsApp API Hooks (Direct Chat Bridge)

2. Core User Portals & Operational Workflows

The system isolates privileges strictly by user role. When an authenticated account logs in, the unified router directs them to their customized experience:

A. Super Admin / Dairy Owner Command Console

1. **Morning Route Batch Generation & Auto-Assignment**
   At 5:00 AM daily, the batch scheduler generates pending delivery tickets for all active subscribers. The system cross-references assigned delivery partners, skips paused vacation accounts, and loads route manifests.

2. **Real-Time Deliveries Monitor**
   Live progress tracking dashboard displaying Total Scheduled Drops, Delivered Count, Pending Drops, and Undelivered exceptions with real-time counters and partner assignment dropdowns.

3. **Fleet Partner Reconciliations & Cash Drawer**
   Driver manager tracking daily cash-on-delivery collections. The owner can review Driver Cash-in-Hand and execute single-click settlements with audit logs.

4. **Dynamic Pricing & Product Catalog Management**
   Full control over per-liter pricing tiers for A2 Cow Milk, Buffalo Milk, Curd, and Desi Ghee with Cloudinary image upload synchronization.

B. Customer Self-Service Portal

1. **Overview & Morning Subscription Health**
   Customers view their daily quota (e.g., 1L Fresh Milk, 250g Dahi), doorstep arrival promise (Before 6:30 AM), assigned driver partner details, and current account balance.

2. **100% Free Vacation Pause & Scheduled Resumption**
   Customers entering out-of-town dates pause their supply with 0 penalty. Deliveries automatically resume on the chosen return date.

3. **Instant 1-Click Delivery Restart**
   When returning early from vacation, customers can restart deliveries immediately via the Overview Plan Card, Delivery Actions, Header Status badge, or Vacation Manager.

4

Next-Day Quantity Temporary Adjustments

Expecting guests or having a function? Customers can temporarily add +1L or +500g Curd for tomorrow morning without altering their base monthly plan.

5

Sidebar-Only Account Controls & Safeguards

Log out and Delete account actions are consolidated strictly within the sidebar footer with modal confirmation safeguards requiring password entry and typing DELETE to confirm.

3. Delivery Partner Operations & Data Schemas

C. Delivery Partner (Driver) Mobile App

1. **Early Morning Shift Launch & Route Mapping**
   Drivers sign in to see their optimized route ordered by apartment and door number, total milk crates required, and total customer count.

2. **Single-Tap Delivery Marking**
   One-click "Mark Delivered" action timestamps the drop and notifies the customer. If unable to deliver, driver selects a reason (Door Locked, Vacation, Customer Refused) which logs to the audit ledger.

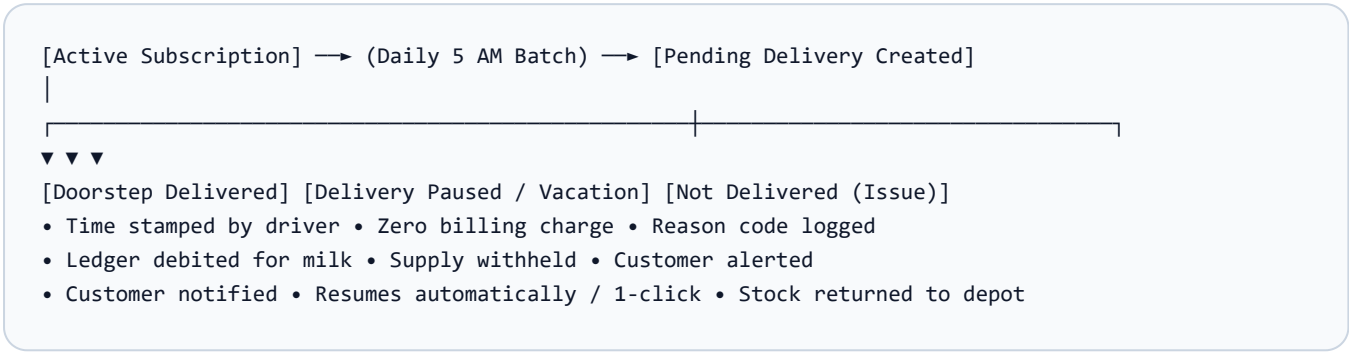
3. **Doorstep Cash Collection Logging**
   Drivers record cash collections on the spot. The customer's outstanding balance drops immediately, and driver's cash-in-hand updates for depot reconciliation.

Database Schema & Data Model Directory

COLLECTION / MODEL	PRIMARY FIELDS & STRUCTURE	KEY RELATIONSHIPS & INDEXING	OPERATIONAL PURPOSE
Customer	`name, mobile, address, area, deliveryPlan (milkQty, curdQty, frequency), currentBalance, status (active/paused), pauseFrom, pauseUntil`	Indexed on `mobile`, `customerId`, `area`	Stores subscriber delivery quota, geographic dispatch routing, and live credit balances.
Delivery	`customerId, customerName, deliveryBoyId, deliveryDate, actualMilk, actualCurd, status (pending/delivered/not_delivered), notDeliveredReason`	Compound index on `deliveryDate + customerId`	Immutable operational delivery tickets generated daily for every doorstep delivery run.
DeliveryBoy	`name, mobile, password, bikeNumber, assignedAreas, activeDeliveriesCount, totalCollectedCash, status`	Indexed on `mobile`	Fleet management, delivery boy assignments, and daily cash drawer reconciliation.
Payment	`customerId, customerName, amount, paymentMethod (cash/upi/bank), receivedBy, date, note`	Indexed on `customerId`, `date`	Digital ledger payment receipts that update customer account balances in real time.
User	`name, email, mobile, password (hashed), role (admin/customer/delivery_boy), status`	Unique index on `mobile`	Stateless authentication, credential verification, and role-based access control.
Product	`title, category, price, volume, unit, description, image, inStock, nutritionalInfo`	Indexed on `category`	Catalog of dairy products (A2 milk, curd, ghee, paneer) for store purchases.
Notification	`recipientRole, recipientId, title, message, type, isRead, createdAt`	Indexed on `recipientRole + recipientId`	Internal alert engine communicating delivery pauses, tickets, and driver assignments.
AuditLog	`action, actor, details, category, timestamp`	Indexed on `timestamp`	Tamper-proof compliance log tracking pricing updates, driver assignments, and deletions.

4. Delivery State Machine & Core REST Endpoints

Delivery Status Lifecycle



Key REST API Directory

METHOD & ROUTE	ACCESS LEVEL	PAYLOAD / PARAMETERS	FUNCTION & RESPONSE OUTCOME
POST /api/auth/login	Public	{ mobile, password }	Validates credentials, returns signed JWT token with user role and profile metadata.
GET /api/deliveries/today	Admin / Driver	Query: ?date=YYYY-MM-DD	Returns consolidated delivery drop manifest with live status and assigned partner.
PUT /api/deliveries/:id	Admin / Driver	{ status, reason, actualQty }	Updates drop status (delivered/not_delivered), records delivery timestamp and reasons.
POST /api/customers/:id/pause	Customer / Admin	{ pauseFrom, pauseUntil, reason }	Sets vacation interval, marks status as paused, cancels pending deliveries in window.
POST /api/customers/:id/resume	Customer / Admin	{ actor }	**Immediately restarts supply:** clears pause flags, reactivates schedule.
POST /api/customers/:id/temp-qty	Customer / Admin	{ date, milkQty, curdQty }	Adjusts tomorrow's delivery quantity without altering base monthly delivery plan.
GET /api/customers/:id/ledger	Customer / Admin	URL Param: :id	Computes gross billing, payments, delivered liter volume, and current outstanding dues.
DELETE /api/customers/me/account	Customer (Self)	{ confirmation: "DELETE", password }	Executes atomic session transaction to purge customer subscriptions, logs, and user auth.

Security & Real-Time Sync Guarantee

The platform enforces strict role isolation (Customers cannot access Admin or Driver tabs; Drivers cannot view owner ledger analytics). Cross-tab synchronization uses browser-level `BroadcastChannel('nmd_realtime_sync')`, ensuring that when a customer restarts their delivery or updates their plan, all open dashboard sessions update without manual page reloads.

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